

BENIKAM SRIKAR

Computer Science Student · Distributed Systems · Visual Computing Research Enthusiast

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EDUCATION

B.Tech in Computer Science and Engineering Aug 2023 – May 2027

Sasi Institute of Technology and Engineering CGPA: 8.45 / 10

Relevant Coursework: Data Structures, Operating Systems, Computer Networks, DBMS, OOP

WORK EXPERIENCE

Data Analytics Virtual Experience Program May 2026

Deloitte Australia (Forage)

Proof of Completion

- Completed Deloitte's Data Analytics Virtual Experience Program involving practical data analysis and forensic technology tasks.
- Analyzed datasets to identify trends, anomalies, and actionable business insights using data-driven methodologies.
- Applied forensic technology concepts to investigate data irregularities and support analytical decision-making.
- Interpreted findings and communicated recommendations through structured business reporting.
- Strengthened analytical thinking, data interpretation, and consulting-oriented problem-solving skills through case-based simulations.

RESEARCH INTERESTS

Distributed Systems, Computer Graphics, Computer Vision, Machine Learning, Visual Computing, Real-Time Systems, Human-Computer Interaction, AI-assisted Content Creation

TECHNICAL SKILLS

Languages	JavaScript, TypeScript, Python, C++
Backend	Node.js, Express, REST APIs, Socket.IO, WebRTC
Frontend	React, Next.js, Tailwind CSS
AI / ML	PyTorch, TensorFlow
Tools & Infra	Docker, AWS, Supabase, Git, Linux

SOFT SKILLS

Communication • Problem Solving • Critical Thinking • Teamwork • Adaptability • Time Management • Group Discussion

CREATIVE TECHNICAL SKILLS

3D Animation	Blender, SideFX Houdini
Video Composition	After Effects, Natron
Concept Art	Photoshop, Midjourney
Video Editing	Premiere Pro, DaVinci Resolve

PROJECTS

SWIFT (Secure Wideband Instant File Transfer) *Solo Project / Live Demo / GitHub*

Real-time file transfer platform with zero persistent storage, live upload-download model, and integrated real-time chat

- Architected a zero-storage file transfer system where transfers happen live between sender and receiver simultaneously unlike WhatsApp, Gmail, or Telegram which store files server-side until the receiver comes online.
- Improved transfer reliability by **40%** and reduced connection setup time by **30%** through iterative architecture redesign across versions.

- Designed a signaling backend using Socket.IO to manage concurrent sessions, peer discovery, and real-time event coordination.
- **v1.0:** Implemented core WebRTC-based single-file transfer using room ID-based peer discovery.
- **v1.2:** Added multi-file transfer with ZIP-based chunking for large payloads (>25MB) and integrated Google OAuth 2.0 authentication.
- **v1.3:** Replaced room-based discovery with email-based host matching, reducing user onboarding friction and simplifying connection flow.
- **v1.4:** Integrated Google Drive fallback for large file transfers (>25MB), increasing successful delivery rate by **60%**.
- **v1.5:** Re-architected transfer model from peer-to-peer WebRTC to a live upload-download pipeline with zero server-side file storage.
- **v1.6:** Implemented real-time chat supporting both group and individual messaging within rooms.

Research and System Design

VEDA — Visual Enhancement and Denoising Assistant

Research-Oriented Project / In Progress

AI-assisted post-processing framework for Blender that enhances Cycles-rendered frames using computer vision and deep learning for professional cinematic quality

- Identified noise and grain artifacts in complex volumetric and dynamic scenes from Blender experimentation and designed an AI-assisted enhancement pipeline.
- Architected a decoupled dual-component architecture consisting of a Blender addon and an independent VEDACore service.
- Designed an automated render-event ingestion pipeline with file-system monitoring and asynchronous processing.
- Implemented cinematic formatting support including Cinemascope and IMAX aspect-ratio adaptation.
- Integrated NAFNet for denoising and Real-ESRGAN for super-resolution and restoration.
- Explored process orchestration, inter-process communication, asynchronous workflows, and GPU inference optimization.
- Demonstrates research potential in Computer Graphics, Computer Vision, Deep Learning, and Operating Systems.

CERTIFICATIONS

OCI Generative AI Professional	Oracle Cloud Infrastructure	2024
OCI AI Foundations	Oracle Cloud Infrastructure	2024
Backend Development & Databases	Coursera	2024
Linux Shell Scripting	Coursera	2024
Deloitte Data Analytics Job Simulation	Forage	2026

ACTIVITIES & INTERESTS

- Built and experimented with real-time systems concepts through hackathon environments, focusing on low-latency communication and event-driven architectures.
- Participated in Google and AMD Hack2Skill hackathons, contributing to prototype systems involving real-time interaction and AI-assisted features.
- Explored internal workings of WebRTC, signaling mechanisms, peer-to-peer networking, and live streaming pipelines through implementation-driven learning.
- Interested in distributed systems design, scalable backend architectures, and production-grade real-time communication systems.
- Actively building intuition around system design trade-offs through projects involving WebSockets, WebRTC, and backend orchestration.